PRISMA 2020 Screening Stage: Deduplication

Report prepared 25 July 2026

1. Input to the Deduplication Step

Step 2 of the screening stage operated on the complete set of 4,798 records identified in Step 1, drawn from all 24 exported RIS files, namely 12 Scopus files and 12 Web of Science (WoS) files. Of these, 22 files report the native output of the Boolean queries and two files, Scopus_manual.ris and WoS_manual.ris, record 18 foundational studies selected manually and added at the identification stage. No filtering had been applied to these records prior to deduplication, so the entire identification-stage pool served as input to this step.

2. Deduplication Method

Records were matched and deduplicated across two sequential passes, in accordance with the SLR screening protocol adopted for this review. In the first pass, records were matched on an exact Digital Object Identifier (DOI), applied case-insensitively and after normalizing https://doi.org/ prefixes together with trailing slashes and extraneous whitespace. In the second pass, applied only to records lacking a DOI, matching proceeded on normalized title and publication year, whereby case, punctuation, and whitespace were standardized and trivial subtitle punctuation differences were disregarded.

Within each identified duplicate group, the most complete record was retained, namely the copy presenting the fullest set of DOI, abstract, journal, volume and page fields. Where completeness was tied between copies, the first-encountered record was retained. All remaining copies within the group were removed.

3. Results

Of the 4,798 input records, 4,734 carried a DOI and were matched into unique DOI groups, from which 1,705 duplicate copies were removed. The remaining 64 records lacked a DOI and were matched by normalized title and publication year, yielding a further four duplicate copies removed. In total, 1,709 duplicate records were removed across both passes, leaving 3,089 unique records surviving Step 2.

3.1 Breakdown by Database

Because the same published study was frequently indexed in both Scopus and WoS, the majority of duplicate copies removed were Scopus copies, with the corresponding WoS copy retained in preference. This pattern held for the manually selected studies as well as the query-

returned records: of the 18 studies recorded in Scopus_manual.ris and WoS_manual.ris, 15 are indexed in both files, and for each of these 15 the Scopus_manual.ris copy was removed as a duplicate and the WoS_manual.ris copy retained, leaving 18 unique manually selected studies among the survivors. Table 1 reports the raw, removed and surviving record counts by source.

Table 1. Deduplication results by database

Database	Raw count (n)	Duplicates removed (n)	Survivors (n)
Scopus (22 query files)	2,894	1,617	1,277
Web of Science (22 query files)	1,871	77	1,794
Scopus manual.ris	18	15	3
WoS manual.ris	15	0	15
Total	**4,798**	**1,709**	**3,089**

3.2 Breakdown by Source File

Table 2 disaggregates the deduplication results to the level of the individual source file, including the two manual files, allowing the contribution of each export to the final surviving pool to be traced. Duplicate removal was concentrated heavily within the Scopus files, consistent with the database-level pattern reported in Section 3.1.

Table 2. Deduplication results by source file

File	Raw count (n)	Duplicates removed (n)	Survivors (n)
Scopus_1.ris	238	108	130
Scopus_3(A).ris	681	283	398
Scopus_3(B).ris	74	48	26
Scopus_3(C).ris	49	39	10
Scopus_4(A).ris	53	40	13
Scopus_4(B).ris	376	259	117
Scopus_4(C).ris	142	109	33
Scopus_5(A).ris	61	44	17
Scopus_5(B).ris	492	409	83
Scopus_5(C).ris	586	225	361
Scopus_6.ris	142	53	89
Scopus_manual.ris	18	15	3
WoS_1.ris	257	0	257
WoS_3(A).ris	297	3	294

File	Raw count (n)	Duplicates removed (n)	Survivors (n)
WoS_3(B).ris	50	0	50
WoS_3(C).ris	49	4	45
WoS_4(A).ris	51	4	47
WoS_4(B).ris	286	6	280
WoS_4(C).ris	112	6	106
WoS_5(A).ris	48	4	44
WoS_5(B).ris	422	8	414
WoS_5(C).ris	245	21	224
WoS_6.ris	54	21	33
WoS_manual.ris	15	0	15
Total (24 files)	**4,798**	**1,709**	**3,089**

This surviving set of 3,089 unique records constitutes the input to Step 3, in which titles and abstracts are screened against the inclusion and exclusion criteria defined for this review. Of the 18 manually selected studies described in the Identification Report, all 18 survive deduplication as unique records: the three indexed in Scopus only survive directly from Scopus_manual.ris, and the 15 indexed in both databases survive as their WoS_manual.ris copy, with the corresponding Scopus_manual.ris copy removed as a duplicate. The complete record of every duplicate pair removed at this step, including these 15, is provided in the accompanying "SLR1_Screening_Stage_Dedup_Audit" workbook.

4. Data Integrity Notes

Only exact duplicate copies were removed at this step, identified strictly through the two matching passes described in Section 2. No records were excluded on the basis of relevance, topical fit, or any other substantive criterion. Consistent with the PRISMA 2020 protocol, deduplication served solely to establish a unique-record pool prior to title and abstract screening, which forms Step 3 of the review pipeline.

5. Document Verification

This report has been prepared, checked, and approved in accordance with the review team's internal quality assurance procedure prior to proceeding to Step 3.

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